

Technical Architecture: AI-Powered Mastery Learning Platform

Companion to the Build Memo — June 2026. Florida B.E.S.T.-aligned, math-first. For technical review.

The one principle everything hangs on

> **The LLM lives in the *authoring* loop, never the *serving* loop.**

When a child is answering a math problem, **no AI call happens**. The questions they see are deterministic variants generated by our own code from pre-vetted templates. The AI is used (a) *offline* to draft those templates, which a human validates before they ever reach a student, and (b) *asynchronously* to diagnose patterns in a student's results and write summaries. This eliminates hallucination at the point of instruction — the failure mode that makes "AI teaches kids" dangerous — while still using AI for the heavy lifting.

This is also how IXL actually works: students "never see the same problem twice," but the underlying **parameterized templates** (structure, answer logic, wrong-answer distractors) are human-authored and fixed. A student answers ~28 variants to reach mastery on a skill; those are randomized instances of a handful of templates, not 28 hand-written questions. We do the same — but use AI to author the templates and to diagnose *why* a student is struggling.

The five layers

1. SKILL GRAPH Florida B.E.S.T. benchmarks → granular skills with prerequisites (skill A unlocks skill B). The map.
2. ITEM BANK Per skill: a few VETTED TEMPLATES (not questions).
(AI-drafted, Each template = structure + parameter ranges + worked human-vetted) solution + common-error distractors. Code generates infinite numeric variants from each template.
3. MASTERY ENGINE Serves variants; tracks a weighted mastery score.
(our code, • Struggling → easier variants / drop to prerequisite deterministic) • Consistent → advance up the graph
Mastery = consistency, not a lucky streak.
4. AI EVALUATOR Reads each student's ERROR PATTERN, diagnoses the (LLM, async, misconception ("repeated sign errors", "setup ok, off answer path) multi-step fails"), assigns targeted remediation, writes parent/teacher summaries. ← the real edge
5. HUMAN LAYER Teacher (school) or parent (home): dashboard, intervention alerts, motivation, safety. Always an adult accountable.

The AI does exactly two jobs, both **off the live answer path**: author templates (offline, human-gated) and evaluate/report (async, over data — never grading a math keystroke in real time).

Layer detail

1. Skill graph

- Built directly on **Florida B.E.S.T.** math benchmarks so the platform is aligned to what the state's **F.A.S.T.** test measures.

- Each grade decomposes into ~150–250 granular skills (math), each with explicit prerequisites. This directed graph drives sequencing and remediation.

2. Item bank (templates, not questions)

- **Per skill: ~3–10 templates.** Each template defines a problem structure, the parameter ranges to randomize, the correct-answer computation, and plausible **distractors tied to specific misconceptions**.
- Code generates unlimited variants → students can't memorize answers; the bank stays small and maintainable.
- **Full Florida math K–8 ≈ a few thousand vetted templates**, not millions of questions. Tractable for one strong teacher + AI over a few months, then maintained weekly.

3. Mastery engine (deterministic, our code)

- Weighted, SmartScore-style score per skill: rewards consistency and difficulty, decays on errors.
- **Within-skill adaptivity:** correct → harder variants; incorrect → easier variants.
- **Between-skill adaptivity:** sustained mastery advances the student; repeated struggle routes to the prerequisite skill.
- Mastery threshold (e.g., sustained high score over N consecutive variants) gates progression — advance on demonstrated mastery, not on time.

4. AI evaluator (the genuine differentiator)

- Runs **asynchronously** over a student's accumulated responses — not in the answer loop.
- Diagnoses *why*, not just *what*: classifies misconception types from error patterns, detects gaming (implausibly fast correct answers, guess patterns), recommends targeted remediation.
- Generates plain-language **summaries for parents/teachers** ("Jayden isn't weak at fractions — he makes one sign error repeatedly; assigned 3 targeted skills").
- Models: cheap model for routine summaries, stronger model for hard diagnostic cases. Student IDs, not names. Zero-data-retention key.

5. Human layer (non-negotiable)

- **Private school:** a teacher circulates, helps, and monitors the live dashboard; intervention alerts surface struggling students.
- **Home/PEP:** the parent sees progress and gets alerts; an optional paid teacher tier provides scheduled help.
- The adult owns motivation, safety, and accountability. The AI never holds sole responsibility for a child.

Assessment strategy

- **Florida F.A.S.T.** (Florida Assessment of Student Thinking): the state's computer-adaptive test, 3×/year (PM1/PM2/PM3), B.E.S.T.-aligned. This is the **external scoreboard** — the Florida analog of MAP. Because our content is B.E.S.T.-aligned, internal mastery data should *predict* F.A.S.T. movement, letting us show parents a projected PM3 gain.
- **For PEP scholarship compliance:** an approved **norm-referenced test** is required annually; plan to support/track it.
- **Internal fast diagnostics:** short 5–10-item adaptive probes built from our bank to place new students, detect regression, and confirm mastery — run continuously, not 3×/year.

Content operations

- **Generate-then-verify pipeline.** AI drafts templates; a hired teacher validates logic, answer keys, and distractors before anything ships. The teacher *validates and extends* — the AI *drafts*. One strong math teacher can vet a large volume of AI-drafted templates quickly.
- **Weekly cadence** to expand coverage and refine weak templates as real student data exposes ambiguity or mis-leveling.

How we beat Khan Academy at math

We do **not** compete on breadth or price (Khan is free and vast). We win on Khan's documented weaknesses:

Khan weakness	Our counter
Gameable — "memorize your way to mastery"; tests pattern-matchable	Randomized variants (no answers to memorize) + consistency requirement + AI gaming-detection
Video-first, passive, low interactivity	Practice-first, immediate feedback; AI explains the student's <i>specific</i> error
Right/wrong only — no diagnosis of <i>why</i>	AI misconception diagnosis — the core differentiator
Weak motivation/competency support	Human accountability + "earn your afternoon" loop + intervention alerts
Generic alignment, content holes	Tight Florida B.E.S.T. alignment → better F.A.S.T. outcomes
Solo; students drift	Dashboards + intervention triggers + an accountable adult

In one line: Khan wins on breadth and free; we win on *ungameable mastery* + *AI that diagnoses why a student is stuck* + *Florida B.E.S.T./F.A.S.T. alignment* + *human accountability* — depth and diagnosis over breadth, for one well-funded market.

What stays in the green zone (recap)

- AI authors content **once, human-vetted**; never improvises live for a child.
- AI **diagnoses and reports**; a human **decides and is accountable**.
- We **own/clean-author** our content; we do not resell another vendor's engine.
- We promise **process and measured growth**, never a guaranteed percentile.

Open technical questions for review

- Skill-graph + mastery-engine: build on a BaaS (Supabase/Postgres) as planned, or is there an existing open mastery-engine worth adopting?
- Template format/DSL: how to represent parameterized templates + distractors so code can generate variants safely and teachers can edit them without engineering help?
- Verification pipeline: how much template QA can be automated (symbolic answer-checking, unit tests per template) vs. requires human eyes?
- Misconception taxonomy: build our own, or adopt an existing math-misconception framework to structure the AI evaluator?
- Cheapest experiment that proves the core bet (AI-drafted + teacher-vetted bank produces real, ungameable mastery) before heavy investment?